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MILITARY SPECIFICATION

* ELECTRODES, WELDING, COVERED, LOW-HYDROGEN, HEAT-TREATABLE STEEL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers low-hydrogen types of arc-welding covered electrodes for use in the manual welding of low alloy steels, which may be heat-treated (quenched and tempered) after welding to develop high physical properties.

* 1.2 Classification. Electrodes shall be furnished in the types and sizes shown in Table I, as specified (see 6.2):

* TABLE I

TYPES AND SIZES

Electrode Material Type	HT - 4130	HT - 4140	HT - 4340	HT - 4340(UHS) (Ultra High Strength)
Position of Welding	All		Horizontal fillets and flats	
Sizes (diameters) (see 3.3.2.1)	1/16, 5/64, 3/32, 1/8, 5/32, 3/16, 1/4			

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitations for bids or request for proposal, form a part of the specification to the extent specified herein.

FSC 3439

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SPECIFICATIONS

Military

MIL-T-5021	Test, Aircraft and Missile Welding Operators Qualification
MIL-S-5626	Steel, Chrome-Molybdenum (4140) Bars, Rods and Forging Stock (For Aerospace Practice)
MIL-I-6868	Inspection Process, Magnetic Particle
MIL-H-6875	Heat Treatment of Steels (Aircraft Practice), Process For
MIL-S-7809	Steel, High Strength, Low-Alloy, Bars, Shapes, Sheet, Strip, and Plate
MIL-W-10430	Welding Rods and Electrodes, Preparation For Delivery of
MIL-S-18728	Steel Plate, Sheet and Strip, Alloy 8630, Aircraft Quality
MIL-S-18729	Steel Plate, Sheet and Strip, Alloy 4130, Aircraft Quality

STANDARDS

Federal

Fed. Test Method Std. No. 151	Metals; Test Methods

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-248	Qualification Tests for Welders (Other Than Aircraft Weldments)
MIL-STD-418	Mechanical Tests for Welded Joints

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STANDARDS

Military (Continued)

MIL-STD-453	Inspection, Radiographic
MIL-STD-779	Reference Radiographs for Steel Welds, Volumes 1 and 2

(Copies of specifications, standards, drawings and other publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

American Society for Testing and Materials Publications

ASTM A370	Standard Methods and Definitions for Mechanical Testing of Steel Products
ASTM E8	Standard Methods of Tension Testing of Metallic Materials
ASTM E23	Standard Methods for Notch Bar Impact Testing of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials 1916 Race Street, Philadelphia, Pennsylvania 19103).

Society of Automotive Engineers Publications

AMS 6358	Steel Sheet, Strip and Plate 0.50 Cr - 0.55 Ni - 0.25Mo (0.38 - 0.43C) (SAE 8740)
AMS 6359	Steel Sheet, Strip and Plate 0.80 Cr - 1.8 Ni - 0.25Mo (0.38 - 0.43C) (SAE 4340)

(Application for copies should be addressed to the Society of Automotive Engineers, 2 Pennsylvania Plaza, New York, N.Y. 10001.)

American Welding Society Publication

AWS 3.0	Welding Terms and Definitions

(Application for copies should be addressed to the American Welding Society, 2501 N.W. 7th Street, Miami, Florida 33125.)

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American National Standards Institute Publication

ANSI Z49.1 Safety in Welding and Cutting

(Application for copies should be addressed either to the American National Standards Institute, 1430 Broadway, New York, N.Y. 10018 or the American Welding Society, 2501 N.W. 7th Street, Miami, Florida 33125.)

3. REQUIREMENTS

* 3.1 First article. The electrodes furnished under this specification shall be a product(s) which has been inspected and has passed the first article inspection specified herein (see 4.5 and 6.2).

* 3.2 Electrodes. The core wire and coverings may be made by any process yielding electrodes conforming to this specification.

3.3 Material.

* 3.3.1 Core wire. The core wires shall be free from piping, segregations, nonmetallic inclusions and other injurious defects.

* 3.3.2 Core wire dimensions and tolerances.

3.3.2.1 Dimensions. Unless otherwise specified (see 6.2) the diameters of the core wire and lengths of electrodes shall be in accordance with the requirements of Table II.

3.3.2.2 Tolerances.

* 3.3.2.2.1 Diameter of core wire. The diameters of core wire shall not vary more than ±0.003 from the nominal diameter specified in the contract or purchase order.

TABLE II

DIAMETERS AND LENGTHS OF ELECTRODES

Nominal Diameter of Core Wire (Inch)	Nominal Length of Electrodes (Inches)	
	End-grip	Center-grip
1/16, 5/64	9	18
3/32	9, 12 and 14	18
1/8, 5/32, 3/16	14 and 15	-
1/4 1/	14 and 18	-

1/ Available only in electrodes for use with 4340 steel

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* 3.3.2.2.2 Length. The lengths of electrodes shall not vary more than ±1/8 inch from the nominal lengths specified in the contract or purchase order.

* 3.3.3 Electrode coverings. Electrode coverings shall be capable of producing satisfactory arc characteristics, shall be consumed uniformly to produce satisfactory production welds, and shall protect the molten metal from contamination. The coverings shall be uniform in quality, adherence, and free from injurious scabs, scales, blisters, pock marks, and other surface imperfections. The covering shall be heat resistant so that it will not flake or crack off when the electrode is used intermittently within the recommended current range for that electrode type and size. The covering shall exhibit no cracking that will adversely affect the operation of the electrode after approximately one-half of the electrode has been consumed or upon resumption of welding.

* 3.3.3.1 Covering composition. Low hydrogen coverings shall be used on all types of electrodes. The chemical composition of the covering shall be optional with manufacturer, except that the total water content (free and combined), shall not exceed 0.2 percent when determined in accordance with 4.8.

3.3.4 Extent of coating.

* 3.3.4.1 Grip ends. Electrodes shall be bare or free from coverings for a distance of approximately 3/4 inch but not to exceed 1-1/2 inches for end grip electrodes, and for approximately 1-1/2 inches but not to exceed 2 inches for center grip electrodes, for making electrical contacts with the holder. The distance from the end of the bare portion to full thickness of the covering shall not exceed 3/4 inch.

* 3.3.4.2 Arc ends. The arc ends of each electrode shall be sufficiently bare and the covering tapered a minimum amount to permit easy striking of the arc. The core wire shall not protrude more than 1/32 inch beyond the covering at the arc end. The covering at the arc end shall show no evidence of chipping for a distance equal to one quarter of the diameter of the core wire or 1/32 inch maximum, whichever is smaller, and shall cover the core wire for at least one quarter of the circumference of the electrode.

* 3.3.4.3 Concentricity. The coverings on all sizes of electrodes shall be concentric with the core to such a degree that the maximum core-plus-one covering dimensions shall not exceed the minimum core-plus-one dimensions by more than 3 percent of the mean dimension at the respective location.

* 3.4 Stability. The coverings shall not be damaged by storage under normal conditions and shall be satisfactory for use after storage for 12 months from date of delivery in original, unopened, hermetically sealed, vapor-proof, metal containers on a dry floor or platform and under a roof.

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* 3.5 Dielectric strength. The coverings at ambient temperatures and in the "dry" condition, that is, as removed from freshly opened containers, shall have a dielectric strength sufficient to insulate effectively against a difference potential of 110 volts, 60 Hz alternating current (ac) or direct current (dc).

* 3.6 Usability characteristics. Electrodes shall be suitable for producing sound dense weld deposits that are reasonably smooth and free from defects such as undercutting, overlapping, lack of fusion, slag inclusion, porosity and cracks, when used with the recommended current and voltage settings, and preheated when required or when specified. Electrodes shall be capable of depositing satisfactory smooth single-pass fillet welds with complete root penetration and shall exhibit complete fusion with the base metal. Fillet welds shall be free from objectionable undercut or overlap.

* 3.6.1 Positioning. Electrodes shall be suitable for depositing weld metal in the positions as indicated in Table I. Welding positions shall be as defined in AWS A3.0. Types HT-4130 and HT-4140 electrodes shall be capable of producing satisfactory welds in the flat, vertical, overhead and horizontal positions. Types HT-4340 and HT-4340(UHS) shall be capable of producing satisfactory welds in the flat and horizontal fillet positions.

* 3.6.2 Slag removal. The slags deposited by the coverings shall be readily removable with hand tools (not air or power operated) from the weld deposits after cooling. Grinding during test plate preparation shall not be used for slag removal and shall be limited to that specified in 3.9.

3.6.3 Fumes. The fumes from the burning coverings during welding shall not be injurious to personnel when electrodes are used in adequately ventilated spaces.

* 3.7 Mechanical properties of weld deposit.

* 3.7.1 Compatibility with base metal. Weld deposited on the indicated base metal (see Table V) shall exhibit mechanical properties that shall be within the limits of Table III after heat treatment when the specimens are prepared and tested in accordance with 4.7.3, 4.7.3.1, 4.7.3.2 and 4.7.3.3.

* 3.7.2 Tensile strength of welded joints. Tensile specimens prepared from weld deposits made with the HT-4130, HT-4140, and HT-4340 electrodes shall be heat treated in accordance with MIL-H-6875. They shall conform to the mechanical properties of Table III, except that yield strength and elongation requirements are not applicable to transverse weld specimens (see 4.7.3.1).

* 3.8 Electrode identification.

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TABLE III

MECHANICAL PROPERTIES OF HEAT TREATABLE ELECTRODES 1/

Electrode Material Types	Mechanical Properties			
	Ultimate Tensile Strength, KSI 2/	Yield Strength, Minimum, 0.2% Offset, KSI	Elongation, 4 X Diameter, Minimum, Percent	Charpy Impact Value, Ft-Lbs, 0°F 3/
HT-4130	125 to 145	100	11.0	47.0
	150 to 170	120	8.0	28.0
	180 to 200	145	6.0	15.0
HT-4140	125 to 145	100	11.0	47.0
	150 to 170	120	8.0	28.0
	180 to 200	145	6.0	15.0
	200 to 220	160	5.0	8.0
HT-4340	150 to 170	120	8.0	25.0
	180 to 200	140	6.0	12.0
	200 to 220	160	5.0	8.0
HT-4340(UHS) 4/	260 to 280	200	4.0	6.0

1/ At no time during the tempering heat treatment shall the actual temperature of the specimen differ from the specified temperature by more than 25°F, when treated as specified in 3.7.2.

2/ Transverse tensile specimens (4.7.3.1 and Figure 3) shall conform to the minimum ultimate tensile strength requirements specified in Table III.

3/ Impact properties shall be determined from V-notch charpy specimens made from material heat-treated in accordance with the requirements of 3.7.1 and 3.7.2.

4/ Tensile specimens for HT-4340(UHS) weld deposit shall be austenitized at 1475-1550°F and tempered at 450°F.

3.8.1 Labeling. The classification number shall be stamped or imprinted on the covering of each electrode at not less than one point or location. An imprinted class designation shall appear as near as practicable to the bare grip-end. The location shall come within the space of 2-1/2 inches from the grip-end of the electrode.

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* 3.8.2 Size and legibility. The imprinted designation shall be composed of equal-size block characters, the height of which shall be approximately 50 to 100 percent of the overall electrode diameter but need not exceed a 5/32 inch height. The imprinted designation shall read from left to right from the grip-end. The color of the print shall contrast with the color of the electrode covering and printed designations shall be readable under normal lighting conditions.

* 3.8.3 Stability. Printed type designations shall remain discernible on electrodes rebaked at temperatures up to 850°F and on unused portions of partially consumed electrodes or discarded stubs, and shall resist effacement by contact incidental to normal handling, shipping and storing.

* 3.9 Grinding. Grinding (or deburring) during welding of a test specimen shall be limited to grinding of weld starts and grinding to correct welder error. Grinding of weld starts shall be limited to the immediate start area (first 1/2 inch of weld bead deposit maximum) only and this shall be done only when considered necessary by the welder. Grinding to correct welder error shall be limited to a maximum of 1 inch of weld bead length for a test specimen. The amount of grinding employed (weld start and welder error) for each test specimen shall be recorded by the welder on the test specimen work sheet record. The record shall indicate the total number of weld starts that were ground and the length and location by weld layer of any grinding operation employed to correct welder error conditions.

* 3.10 Workmanship. Electrodes shall be manufactured in accordance with high-grade commercial practice covering the class of work. The electrodes shall be uniform in quality, clean, sound and smooth. The material shall be free from injurious scabs, blisters, pock marks, bruises, or other surface defects. The electrodes shall be free of any materials, conditions, or defects which will affect adversely the operation of the electrodes, or which will create unsatisfactory operating characteristics or unsound weld metal.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Lot. An inspection lot shall consist of electrodes of the same type and size produced under the same conditions from the same master heat of steel for wire and the same batch of coating material, packaged as specified in the contract or order and offered for inspection at one time. Unless otherwise specified in the contract or order, the maximum weight of a lot shall be 6000 pounds.

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* 4.3 Sampling. Unless otherwise specified, sampling plans and procedures for inspection shall be in accordance with the provisions set forth in MIL-STD-105.

* 4.4 Classification of tests. The inspection and testing of electrodes shall be classified as follows:

a. First article inspection (see 4.5)

b. Quality conformance inspection (see 4.6)

* 4.5 First article inspection. First article inspection shall consist of all inspection procedures of this specification (see 4.6.2, 4.7 and 4.8). The responsibility for performance of first article inspection shall be specified by the procuring activity (see 6.2).

* 4.5.1 First article samples. First article test samples shall be obtained from the material to be furnished for each type. Samples shall be selected at random from electrodes which have been manufactured using the same production processes, procedures and equipment which will be used in filling the contract. The samples shall be selected from either filled unit containers or from the production line immediately prior to packaging. If selected from the production line, the total sample shall be selected throughout the run so all parts of the run are represented. To verify compliance for visual and dimensional examination, as well as compliance with Section 5, the test samples shall be selected in accordance with MIL-STD-105, Inspection Level II, Acceptable Quality Level (AQL) of 2.5 percent defective. For all other tests, a random sample of sufficient quantity shall be selected in accordance with MIL-STD-105, Inspection Level S-1, Acceptance Number of zero.

* 4.5.2 Data to accompany test samples. Whether or not first article samples are required (see 4.5.3), the manufacturer shall supply a certified statement that the welding electrodes comply with the requirements of 3.3.1 and 3.4 as well as any restrictions as applicable to usable welding positions (see Table I) and welding processes. The certificate shall also contain actual quantitative test results indicating the process and range of amperage employed as well as that of grinding (see 3.9) in making the required usability tests for each lot.

* 4.5.3 Prior approval. If the manufacturer has previously delivered a material of a specified type that conforms to all requirements of this specification and his material has been found to be satisfactory, the requirements for first article inspection for that product, in connection with any subsequent contract or order for that material, may be waived at the discretion of the procuring activity (see 6.2.3.1).

* 4.5.4 Further production. Any production of electrodes by the contractor prior to approval by the procuring activity shall be at the contractor's risk.

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* 4.6 Quality conformance inspection.

* 4.6.1 Sampling.

* 4.6.1.1 For filled containers. Samples for examination of filled containers, both shipping containers filled with unit containers and filled unit containers, shall be selected at random from each lot (see 4.2) in accordance with MIL-STD-105, Inspection Level II, Acceptable Quality Level 2.5 percent defective.

* 4.6.1.2 For visual and dimensional examinations. Samples for visual examination and for dimensional examination shall be selected at random from each lot (see 4.2) in accordance with MIL-STD-105, Inspection Level II, Acceptable Quality Level 2.5 percent defective. The material selected at random for this sample shall be taken from packages selected in the packaging inspection sample (see 4.6.1.1) or from the production line (see 4.5.1).

* 4.6.1.3 For other tests. One or more unit package of electrodes shall be taken at random, from the unit packages selected from each lot in accordance with 4.6.1.1, or random samples of electrodes shall be selected from the production line in accordance with MIL-STD-105, Inspection Level S-1, Acceptance Number of zero. The samples of electrodes selected shall be subjected to the tests specified in Table IV.

TABLE IV

OTHER TESTS

Tests	Paragraphs Giving Further Reference	Paragraphs Giving Requirements
Concentricity	4.6.2.5	3.3.4.3
Dielectric strength	4.6.2.6	3.5
Usability characteristics	4.7.1.4	3.3.3 and 3.6
Fillet weld test	4.7.2.2	3.6

* 4.6.2 Quality conformance examination and tests.

* 4.6.2.1 Containers. Each of the filled shipping containers and each unit package selected in accordance with 4.6.1.1 shall be examined for evidence of parted seams and other construction defects, for unsatisfactory markings, for closure, and for weight of contents to determine conformance with Section 5. Any rigid metal container with one or more defects shall be rejected. Unit packages shall be subjected to the leakage test in accordance with 4.6.2.2.

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* 4.6.2.2 Leakage test (unit packages). Sample filled unit packages selected in accordance with 4.6.1.1 shall be immersed in water at 150°F or above for 2 minutes. The presence of leaks, as evidenced by the emanation of air bubbles or otherwise, shall be cause for rejection of the package and contents. When defective packages are found in a shipment, the shipment represented shall be rejected or, at the option of the contractor, all unit packages shall be tested and accepted or rejected on an individual basis.

* 4.6.2.3 Visual examination. Sample electrodes, selected in accordance with 4.6.1.2 shall be visually examined for condition of the covering and obvious defects to determine conformance to 3.3.3 and 3.3.4, for identification to determine conformance to 3.8, and for workmanship to determine conformance to 3.10.

* 4.6.2.4 Dimensional examination. The sample electrodes selected in accordance with 4.6.1.2 shall be measured for dimensions to determine conformance to 3.3.2, 3.3.4.1 and 3.3.4.2. The diameter shall be measured using standard micrometer methods or go-no-go gages for the uncovered portions of the core.

* 4.6.2.5 Concentricity of covering. Five or more electrodes from each sample unit package selected in accordance with 4.6.1.3 shall be tested for concentricity of covering. The core wire shall be exposed by removing a small amount of covering from a spot on one side of the core wire near one end of the covered length, care being taken to insure that no metal is removed from the core wire. The diameter of the core wire and the thickness of the covering on the side opposite the bared spot shall be measured with a micrometer. The covering shall then be removed from a spot on the opposite side of the core wire at a point immediately adjacent to that at which the first measurement was made, and a similar measurement made. Second and third pairs of similar measurements shall be made at approximately mid-length and at the other end of the covered portion, or diameters approximately 60 degrees and 120 degrees, respectively, from the diameter on which the first pair of measurements were made. A pair of measurements (two adjacent measurements) which show the greatest variation shall be used to determine the acceptability of the electrodes (see 3.3.4.3).

* 4.6.2.6 Dielectric strength. Two or more electrodes from each sample unit package selected in accordance with 4.6.1.3 shall be tested by laying each electrode flat on a steel plate and applying a potential of 110 volts dc between the grip end of the electrode and the steel plate. If the electrode covering effectively prevents the flow of current, that is, if no arcing occurs, the dielectric properties shall be considered satisfactory (see 3.5).

* 4.6.2.7 Stability and welding characteristics. The manufacturer shall submit a notorized certification signed by a responsible official of its management attesting to the effect that the electrodes are the same as that complied with by first article inspection (see 4.5) and will meet all requirements of this specification as specified in 3.4 for stability, in 3.6 for usability, in 3.7 for mechanical

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properties of weld deposition, and in 3.3.3.1 for covering composition. Any restrictive positioning (see 3.6.1) and processing shall also be stated.

* 4.7 Welding tests.

* 4.7.1 Conditions applicable to welded specimens.

* 4.7.1.1 Welders. Welding of test joints shall be accomplished by welders qualified in accordance with specifications or standards applicable to the procuring activity (see MIL-T-5021 and MIL-STD-248).

* 4.7.1.2 Materials. Sheet and plate materials used for specimens to be welded with the HT types of electrodes shall be in accordance with Table V.

* 4.7.1.3 Preheating and post heating. Use of preheating and post heating shall be optional. However, it is recommended that sheets and plates to be used in the preparation of welded test specimens should be preheated and post heated in accordance with the schedules stated in Table V.

* 4.7.1.4 Usability characteristics. The operating characteristics of the electrodes and the burning characteristics for conformance to 3.3.3 and 3.6 shall be observed during fabrication of welded test specimens. Welds shall be examined externally by visual inspection for smoothness of surface, surface porosity, cracking, undercutting and overlapping. Welds shall be examined for the presence of internal defects by visual inspection of the fractured surfaces of transverse weld tensile specimens after fracture in tension. Test plates welded as specified in 4.7.3.2 shall be examined magnetically and radiographically in accordance with MIL-I-6868 and MIL-STD-453 respectively after removal of the weld reinforcement and backing plate by machining flush with the adjacent plate surfaces. Magnetic indications definitely established as cracks shall be cause for rejection. Indications of defects exceeding those shown in MIL-STD-779, Volumes I and II for Grade 1, Fine Scattered Porosity in size and frequency shall be cause for rejection of welded test plates 1/4 inch or less in thickness and 3/8 inch or greater in thickness, respectively.

* 4.7.2 Fillet weld test. Fillet welded T-joint assemblies, conforming to Figures 1 or 2, as applicable, shall be welded in the horizontal position only. Materials shall be in accordance with 4.7.1.2 and Table V, using the indicated base metal thickness for the electrode under test. Test plates may be preheated and postheated in accordance with 4.7.1.3 and the schedule indicated in Table V. Two specimens shall be welded. When testing any 3/16-inch diameter electrode, an initial root pass with a smaller diameter electrode of the same type and manufacture is permissible.

* 4.7.2.1 First article inspection. Double fillet T-joint assemblies, conforming to Figure 1 shall be prepared. The weld shall be visually examined

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for usability characteristics (see 4.7.1.4) to determine conformance to 3.6. The weld shall not show excessive convexity or concavity. At a point approximately 1 inch back from the crater end of the deposit made with the first electrode (on the first fillet), the specimen shall be sectioned. One surface of each section shall be polished and examined for cracking, gas pockets, incomplete root fusion or presence of weld defects. Root fusion shall be complete, except that unfused areas of maximum width less than 10 percent of the theoretical leg of the fillet will be permitted. The presence of any of these defects shall be cause for rejection.

TABLE V

SHEET AND PLATE MATERIALS

Electrode Material Type	Sheet and Plate, Conforming to	Preheating and Postheating Schedules	
		Preheat - °F	Postheat - °F
HT-4130	MIL-S-18728 or MIL-S-18729	200	300 for 30 minutes
HT4140	MIL-S-5626 or AMS 6358	550 to 600	550 to 600 for 30 minutes
HT-4340 and HT-4340(UHS)	AMS 6359	550 to 600	550 to 600 for 30 minutes

* 4.7.2.2 Quality conformance inspection. Single fillet T-joint assemblies conforming to Figure 2 shall be prepared. The weld shall be visually examined for usability characteristics (see 4.7.1.4) to determine conformance to 3.6. The joint shall then be tested by bending in the direction indicated in the figure until the weld has fractured. The fracture surface shall then be examined for the presence of root defects to determine conformance to 3.6. Root fusion shall be complete, except that unfused areas of aggregate length less than 10 percent of the weld, and of maximum width less than 10 percent of the theoretical leg of the fillet, will be permitted.

* 4.7.3 Tensile tests. The type of tensile test, thickness of the base and number of specimens required to determine conformance to 3.7.1 and 3.7.2 shall be as shown in Table VI.

* 4.7.3.1 Transverse tensile specimens. Test plates shall be prepared in accordance with Figure 3 and the notes applied thereto from materials in accordance with 4.7.1.2 and Table V. Test plates shall be welded on one side only in the flat position as defined in AWS 3.0 using the indicated base metal thickness for the

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electrode under test. Test plates may be preheated and postheated in accordance with 4.7.1.3 and the schedule indicated in Table V. After welding, the test plates shall be heat treated in accordance with 3.7.2 to conform to the notes of Table III. Tensile specimens shall be cut from the locations indicated by Figure 3.

TABLE VI

TENSILE TESTS

Applicable Tensile Tests	Nominal Diameter of Electrode-Inch (See 3.3.2.1)						
	1/16	5/64	3/32	1/8	5/32	3/16	1/4
Transverse Tensile (See 4.7.3.1) Nominal Base Metal Thickness-Inch	1/16	1/16	1/8	1/4	1/2	1/2	3/4
Number of Specimens for each strength level	3	3	3	3	3	3	3
All-Weld-Metal Tensile (See 4.7.3.2) Nominal Base Metal Thickness-Inch	-	-	-	1/2	3/4	3/4	3/4
Number of Specimens for each strength level	-	-	-	4	4	4	4

* 4.7.3.2 All-weld-metal tensile specimens. Test plates shall be prepared in accordance with Figure 4 and the notes thereto from materials in accordance with 4.7.1.2 and Table V. The test plates shall be welded on one side only in the flat position as defined in AWS 3.0. The beveled edges of the base metal plates and the surfaces of the backing plate shall be clad with a layer of weld metal (not less than 3/16 inch thick) deposited with the electrode being tested. The plate edges shall be clad, using a stringer bead technique and machined or ground smooth before assembly of the test plate. The plate shall be insulated from the bench by 1/2 inch of asbestos during welding. Plates may be preheated and postheated in accordance with 4.7.1.3 and the schedule indicated in Table V. After welding the test plates shall be heat treated in accordance with 3.7.2 to conform to the requirements specified in the notes of Table III prior to testing.

* 4.7.3.2.1 Welding techniques. Welds shall be deposited in layers, using the 2-1/2 diameter split weave buildup shown in Figure 4. Layers shall be approximately 1/8 inch thick and shall be deposited in such a manner that the final layers include a reinforcement of standard proportions (1/32 to 1/8 inch).

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* 4.7.3.2.2 Machining of tensile specimens. The all-weld-metal tensile specimens shall be machined from locations indicated in Figure 4. All specimens shall be machined 0.020 inch oversize prior to heat treatment. Heat treated specimens shall be finished to the final dimensions for round tension specimens R1 as detailed in MIL-STD-418 when testing electrodes 5/32 inch or larger and R3 as detailed in MIL-STD-418 when testing electrodes 1/8 inch or less.

* 4.7.3.3 Testing of tensile specimens. Tests of specimens prepared in accordance with 4.7.3.1 and 4.7.3.2 shall be conducted in accordance with the applicable requirement of ASTM E8.

* 4.7.4 Impact test. The impact properties of welded specimens shall be determined by tests, in triplicate, of charpy V-notched specimens, heat treated to each strength level of the respective type electrode for conformance to 3.7 and Table III.

* 4.7.4.1 Preparation of specimens. The impact specimens shall be prepared in accordance with ASTM E23. The test plates shall be prepared as shown in Figure 5 from plate material conforming to MIL-S-7809. The beveled edges of the base metal plates and the surface of the backing plate shall be clad with a layer of weld metal (not less than 3/16 inch thick) deposited with the electrode being tested. The plate edges shall be clad, using a stringer bead technique and machined or ground smooth before assembly of the test plate. The plate shall be insulated from the bench by 1/2 inch of asbestos during welding. Plates may be preheated and postheated in accordance with 4.7.1.3 and the schedule indicated in Table V. After welding, the test plates shall be heat treated in accordance with 3.7.2 and Table III prior to testing.

* 4.7.4.2 Machining. Specimens shall be machined from the locations shown in Figure 5. Unnotched specimen blanks shall be machined 0.020 inch oversize prior to heat treatment. After heat treatment, the impact specimens shall be finished machined to the final dimensions and notch inserted as indicated in Figure 11 of of ASTM A370 for simple beam charpy V-notched impact test specimens. Precautions should be observed during machining of the notch, so that no excessive heat is generated which might affect the structural properties of the weld metal.

* 4.7.4.3 Method of tests. Testing of the specimens shall be conducted at 0°F. The three impact values obtained shall be averaged and no individual specimens shall show an impact value which varies from the average by more than 25 percent. The weld metal shall be considered unsatisfactory if the average impact value does not conform to Table III. A second set of specimens shall be retested if the first set fails to conform to this specification.

4.8 Moisture Test. The tendency for the electrode coatings to generate gas (especially hydrogen), which may be dissolved in the molten weld metal and evolved after solidification of the weld metal, shall be determined by analysis of the coating of one electrode of each size for total water content

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(hygroscopic and combined). Electrodes shall be considered unacceptable unless the total water content is in conformity with 3.3.3.1.

4.8.1 Apparatus for determining moisture content. The apparatus consists essentially of the following component parts:

a. A purifying and drying train using an ascarite tower, a gas wash bottle containing 90 percent concentrated sulfuric acid by volume, and a U-tube filled with dehydrite (magnesium perchlorate).

b. A heating chamber consisting of a high temperature combustion tube, 1 inch diameter by 30 inches long mounted in a high-temperature furnace (Globar type or induction) equipped with temperature indicating instruments and controls.

c. An absorption train made by using a glass wool filter, two U-tubes containing dehydrite, and a gas wash bottle containing concentrated sulfuric acid. A schematic sketch showing the arrangement of the apparatus is shown in Figure 6.

4.8.2 Operation of equipment. In operation, the train is swept by a stream of pure oxygen which passes over the sample of covering contained in a boat (approximately 9/16 by 13/16 by 4-1/4 inches) which has been heated to 1,800°F. The water driven from the sample is collected in the absorption train and the quantity is determined by the increase in weight of the dehydrite at the absorption end of the train.

4.8.3 Analytical procedure. The following procedure shall be employed.

4.8.3.1 Preparation of sample. An electrode shall be taken from a freshly opened container, and a sample of the coating obtained by quickly removing it from the electrode by bending or hammering. The chips shall be transferred immediately to a vial and sealed with a rubber stopper until ready for use.

4.8.3.2 Ignition of combustion tube and boat. Before the train is used for analysis, the water content of the combustion tube and boat must be driven off. This is accomplished by placing the boat in the center of the combustion tube, assembling the train, sweeping with oxygen at a rate of 200-225 milliliters (ml) per minute and bringing the combustion tube to a temperature of 1,800°F. in a heating period of not less than 2 hours. Hold at temperature (1,800°F.) for 2 hours. At the end of the 4-hour heating cycle, place weighed dehydrite U-tubes in the absorption end of train and continue operation at 1,800°F. for 1 hour. Remove U-tubes, place in dessicator for 20 minutes, after which reweigh and determine gain in weight.

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A weight gain not in excess of 0.0015 gram shall be considered a satisfactory blank determination, indicating the train to be ready for use. Otherwise, the ignition shall be continued until a satisfactory blank determination is obtained. When satisfactory, the ignited boat is removed from the combustion tube, cooled in air to about 350°F., placed in a dessicator and cooled to room temperature. The combustion tube shall be closed and maintained at 1,800°F.

4.8.3.3 Determination of water content of sample. The combustion tube and boat having been satisfactorily ignited, the boat cooled, the furnace operating uninterrupted at 1,800°F., the oxygen flowing, and the tared dehydrite absorption tubes in place, the apparatus is ready for use. Quickly weigh a 1 to 2-gram sample of electrode coating (large chip or chips), rapidly transfer to ignited and cooled boat, immediately place in center of combusion tube, and close train. Operate for 1 hour at temperature (1,800°F.). At end of cycle, remove absorption tubes, determine gain in weight, and calculate percent of water removed from sample at 1,800°F.

4.9 Rejection and retest.

4.9.1 Rejection. When electrodes or specimens welded with electrodes fail to meet any of the tests required by this specification, the lot represented shall be subject to rejection, except as otherwise provided by MIL-STD-105. Retests shall be permitted in accordance with the requirements of Fed. Std. No. 151.

4.9.2 Resubmitted lots. Paragraph 6.4 of MIL-STD-105 shall apply except that a resubmitted inspection lot shall be inspected using tightened inspection. For visual examination where the original acceptance number was zero, a sample size represented by the next higher sample size code letter shall be chosen.

5. PREPARATION FOR DELIVERY

* 5.1 Packaging. The levels of packaging shall be Level A or Level C, unless otherwise specified (see 6.2).

* 5.1.1 Level A. Unless otherwise specified in the contract or order, the moisture sensitive electrodes shall be packaged in accordance with MIL-W-10430. A unit package shall contain one type and size as specified by the procuring activity. Electrodes shall be unit packaged in 5, 10, 25, or 50-pound quantities net weight as specified. Unit containers shall be Class 1a, 1b or 1c of MIL-W-10430.

* 5.1.2 Level C. The electrodes shall be packaged in accordance with Level C requirements of MIL-W-10430 to afford the minimum degree of protection necessary to prevent deterioration or damage during shipment under normal environmental conditions and commercial modes of transportation.

* 5.2 Packing. Packing shall be Level A, Level B or Level C as specified (see 6.2).

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* 5.2.1 Level A. Electrodes, packaged as specified, shall be packed for oversea shipment in accordance with MIL-W-10430.

* 5.2.2 Level B. Electrodes, packaged as specified, shall be packed for domestic shipment in accordance with MIL-W-10430.

* 5.2.3 Level C. Electrodes that require overpacking by the carrier shall be packed in exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery and shall meet as a minimum, the requirements of the rules and regulations applicable to the mode of transportation selected.

* 5.3 Marking. In addition to any special markings required by the contract or order, interior packages, exterior shipping containers and palletized unit loads shall be marked in accordance with MIL-W-10430 and MIL-STD-129.

* 5.3.1 Unit containers. Unit containers shall bear a printed label in a conspicuous location on or near the manufacturer's label and on any special instructions which may be enclosed as follows:

WARNING

Welding may produce fumes and gases hazardous to health. Avoid breathing these fumes and gases. Use adequate ventilation. See American National Standard Z49.1, Safety in Welding and Cutting.

* 5.4 Shipping container tests. Tests on shipping containers as specified in MIL-W-10430 shall be applied to Levels A and B packing only.

6. NOTES

6.1 Intended use.

6.1.1 HT-4130. The HT-4130 electrode is intended for use in the welding of 4130 or 8630 steels which will be heat-treated to a tensile strength of 125,000 pounds per square inch (psi) to 200,000 psi.

6.1.2 HT-4140. The HT-4140 electrode is intended for use in the welding of 4140 or 8740 steels which will be heat-treated to a tensile strength of 125,000 psi to 220,000 psi.

6.1.3 HT-4340. The HT-4340 electrode is intended for use in the welding of 4340 steel which will be heat-treated to a tensile strength of 150,000 psi to 220,000 psi.

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6.1.4 HT-4340 (UHS). The HT-4340 (UHS) electrode is intended for use in the welding of 4340 steel which will be heat-treated to a tensile strength of 260,000 psi to 280,000 psi.

6.2 *Ordering data.*

6.2.1 Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Type of electrode required (see 1.2 and Table I).

c. Dimensions of electrode required (see 1.2, 3.3.2.1 and Table II).

d. Chemical composition of weld metal deposit, if other than specified (see 3.8).

e. Lot size (see 4.2).

f. Responsibility for performance of First Article inspection (see 4.5).

g. Levels of packaging and packing required (see 5.1 and 5.2).

h. Additional markings, if required (see 5.3).

6.2.2 Electrodes shall be ordered by the pound.

6.2.3 Contracts or orders shall specify the following provisions for First Article inspection.

6.2.3.1 Whether First Article inspection is required. When a manufacturer is in continuous production of electrodes from contract to contract, consideration should be given to waive the First Article inspections. If inspection is required, indicate:

a. Where the First Article inspection is to be conducted (at the manufacturer's plant or Government or commercial laboratory).

b. That the approval of First Article samples or the waiving of the First Article inspection shall not relieve the manufacturer of his obligation to fulfill all other requirements of the specifications and contract.

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6.3 Data. For the information of manufacturer and Contracting Officers, any of the data specified in (a) subparagraphs below, (b) applicable documents listed in Section 2 of this specification, or (c) referenced lower-tier documents need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order. The data to be furnished shall be listed on DD Form 1423 (Contractor Data Requirement List), which shall be attached to and made a part of the contract or order.

6.3.1 First Article data. When First Article samples are submitted (see 6.2), they should be accompanied by a complete inspection report showing the results of the manufacturer's inspections. The inspection shall include the following:

a. Report of inspections, together with a detailed statement indicating compliance or extent of noncompliance with all requirements of this specification, referring specifically to paragraph numbers. Whenever a requirement is considered to be not applicable, the report shall so state.

b. A complete description of inspection equipment and inspection procedures.

c. Reproducible outline and description conditions. Where inspections specified in this specification are not considered applicable, the reason, and the substituted inspection should be clearly described.

d. Copies of inspection log sheets.

6.4 Precautionary notes.

* 6.4.1 Combination of materials. Electrodes having first article approval for use in welding specific alloy steels may or may not be capable of producing satisfactory welds in other types of steels or combinations thereof.

* 6.4.2 Preheating and postheating of production parts. When production parts are to be welded with electrodes which are capable of developing tensile strengths of 150 KSI or above when subjected to appropriate heat treatment, the electrode manufacturer's recommendations concerning preheat and postheat treatment of weldments should be observed. In the absence of instructions to the contrary, parts fabricated from 4130 and 8630 steels should be preheated and post treated as stated in Table V for HT-4130 electrode. Parts fabricated from 4140, 4340, or 8740 steels should be preheated and post treated as stated in Table V for HT-4140 and HT-4340 electrodes. Heating should be continued for not less than 10 minutes after completion of welds in sheets less than 1/4 inch, and for not less than 30 minutes after completion of joints in plates 1/4 inch or heavier. Preheat temperature should not exceed 800°F at any time.

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* 6.5 Marginal indicia. The margins of this specification have been marked to indicate where changes, deletions, or additions to the previous issue have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these documents. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the marginal notations and relationship to the last previous issue.

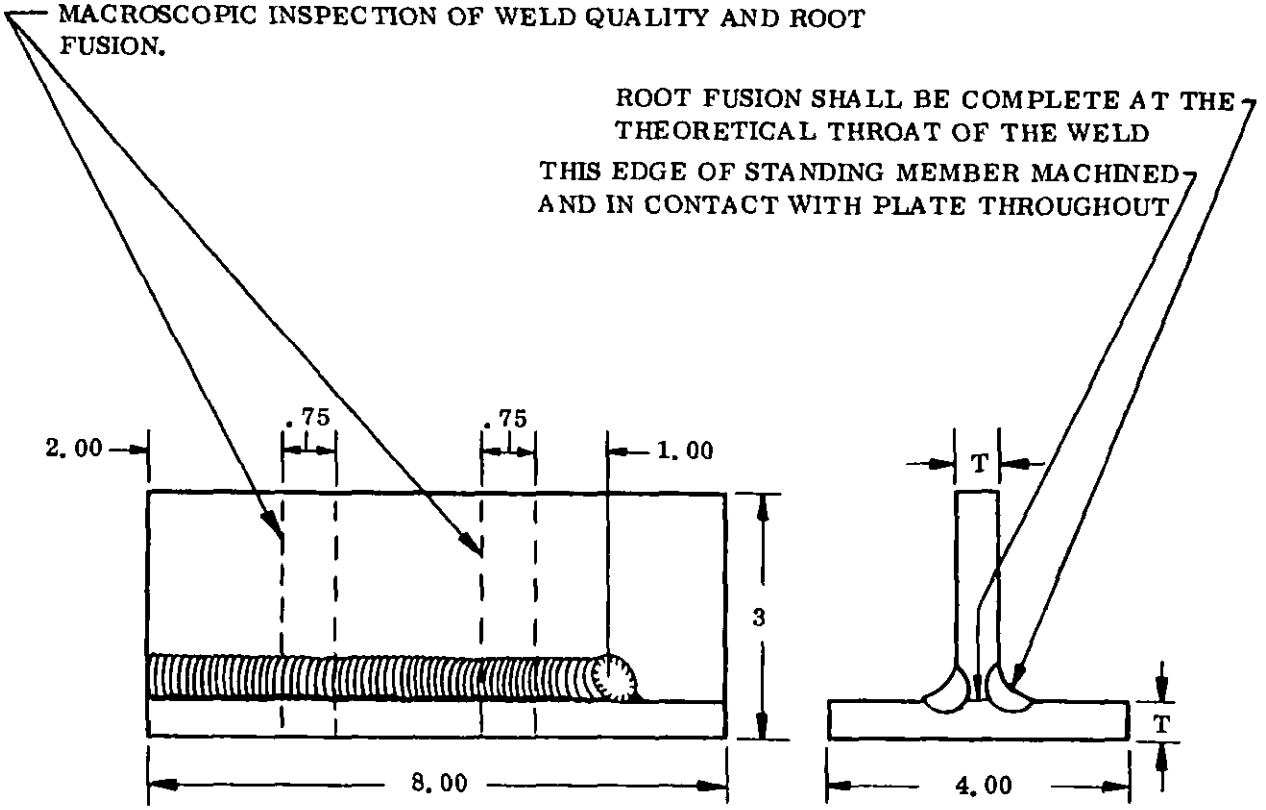
Custodians:
Army - WC
Navy - AS
Air Force - 11

Review activity:
Army - None
Navy - None
Air Force - 84

User activity:
Army - AT, MU
Navy - SH
Air Force - None

Preparing Activity:
Navy - AS
Project No. 3439-0094

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ELECTRODE SIZE	THICKNESS (T)	FILLET WELD SIZE
5/64	1/8	1/8
3/32	1/4	3/16
1/8	3/8	1/4
5/32	3/8	5/16
3/16	1/2	3/8
1/4	1/2	7/16

ALL ELECTRODES SHALL BE TESTED IN THE HORIZONTAL POSITION. PLATE MATERIAL SHALL BE AS DETAILED IN 4.7.1.2 AND TABLE V.

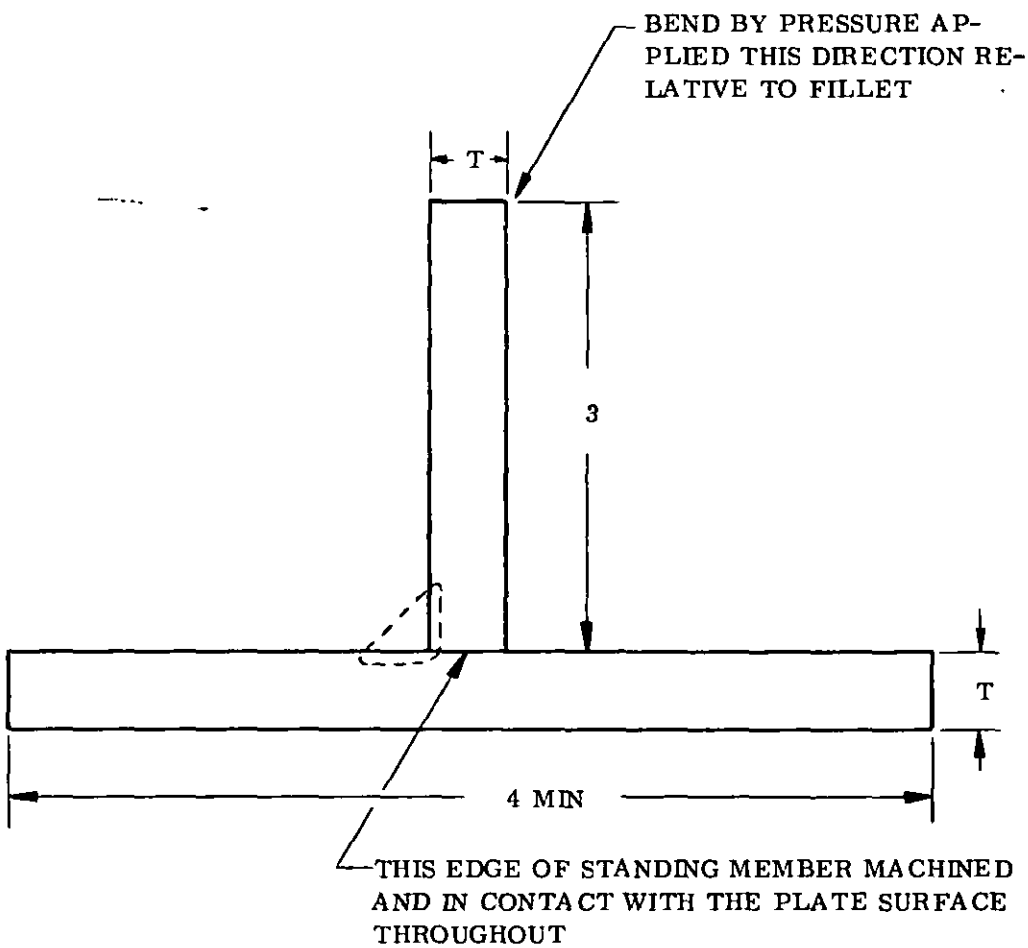
AFTER MAKING ONE FILLET, COOL TO ROOM TEMPERATURE BEFORE MAKING FILLET ON OPPOSITE SIDE.

USE PREHEAT AND POSTHEAT AS RECOMMENDED.

DIMENSIONS IN INCHES.

FIGURE 1. FILLET WELD SPECIMENS

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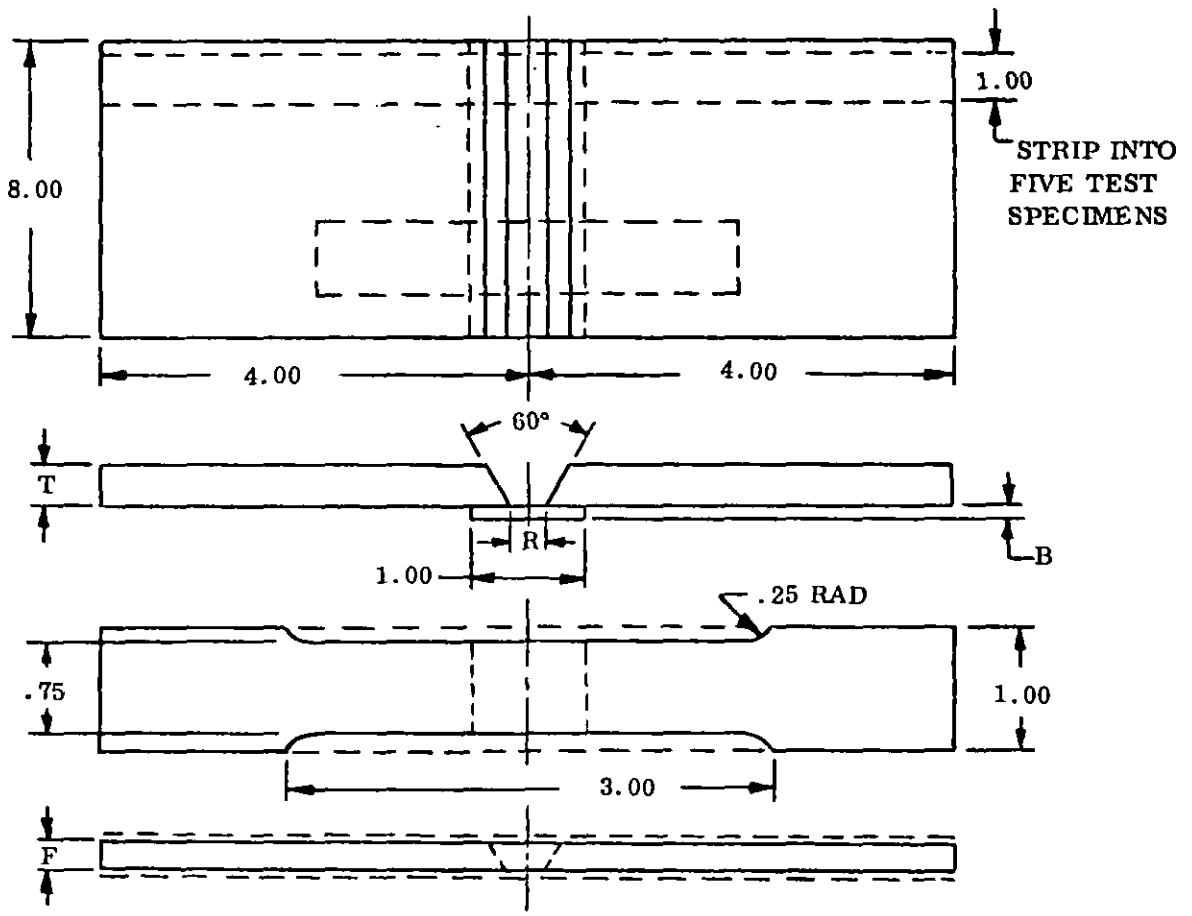


SEE FIGURE 1 FOR TABLE AND NOTES.

DIMENSIONS IN INCHES.

FIGURE 2. SINGLE FILLET WELD SPECIMEN

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DIMENSIONS FOR MECHANICAL PROPERTIES TENSILE SPECIMENS

DIMENSIONS FOR	SIZE-ELECTRODES				
	1/16 & 5/64	3/32	1/8	5/32 & 3/16	1/4
PLATE THICKNESS (T)	1/16	1/8	1/4	1/2	3/4
GROOVE WELD OPENING (R) AND BACKING STRIP THICKNESS (B)	1/16	3/32	1/8	3/16	1/4
FINISHED SPECIMEN THICKNESS (F)	0.04	0.10	0.20	0.40	0.60

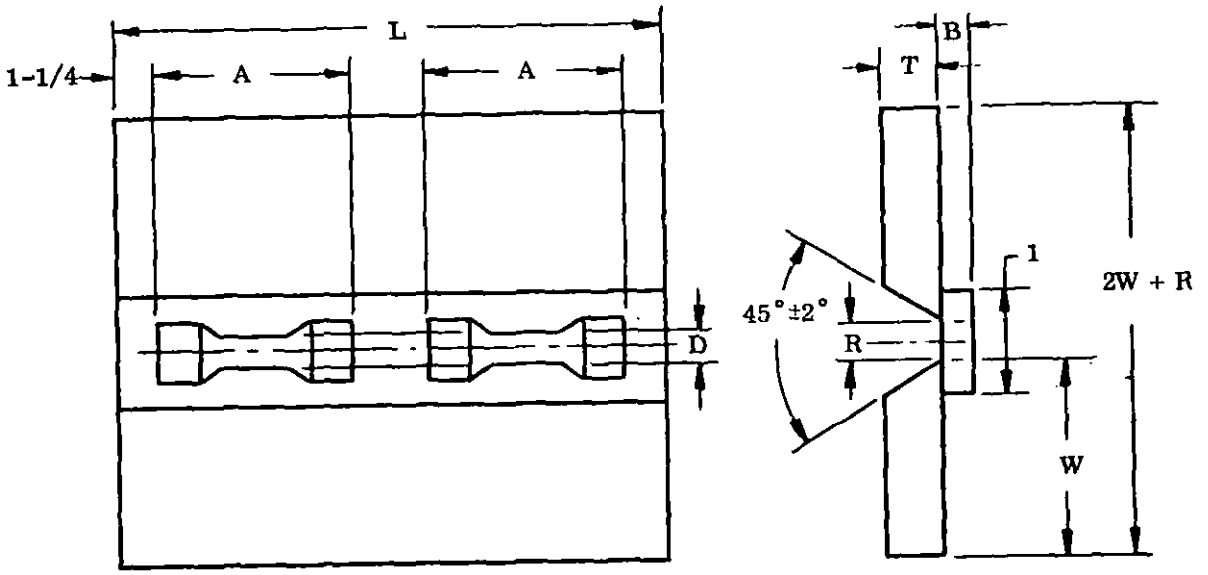
SPECIMENS SHALL BE TESTED IN THE CONDITIONS SPECIFIED IN TABLE III FOR THE TYPE BEING TESTED.

THE MINIMUM MECHANICAL PROPERTIES OF TENSILE TEST SPECIMENS SHALL BE NOT LESS THAN THE VALUES SPECIFIED IN TABLE III FOR THE TYPE OF ELECTRODE BEING TESTED.

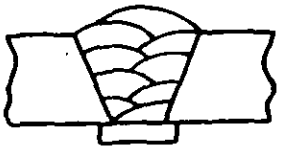
DIMENSIONS IN INCHES.

FIGURE 3. TRANSVERSE TENSILE SPECIMEN

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TEST PLATE FOR CHECKING USABILITY CHARACTERISTICS OF ELECTRODES AND PROPERTIES OF WELD DEPOSITS.
BACKING STRIP SHALL BE OF THE SAME NOMINAL COMPOSITION AS BASE METAL.
DIMENSIONS IN INCHES.

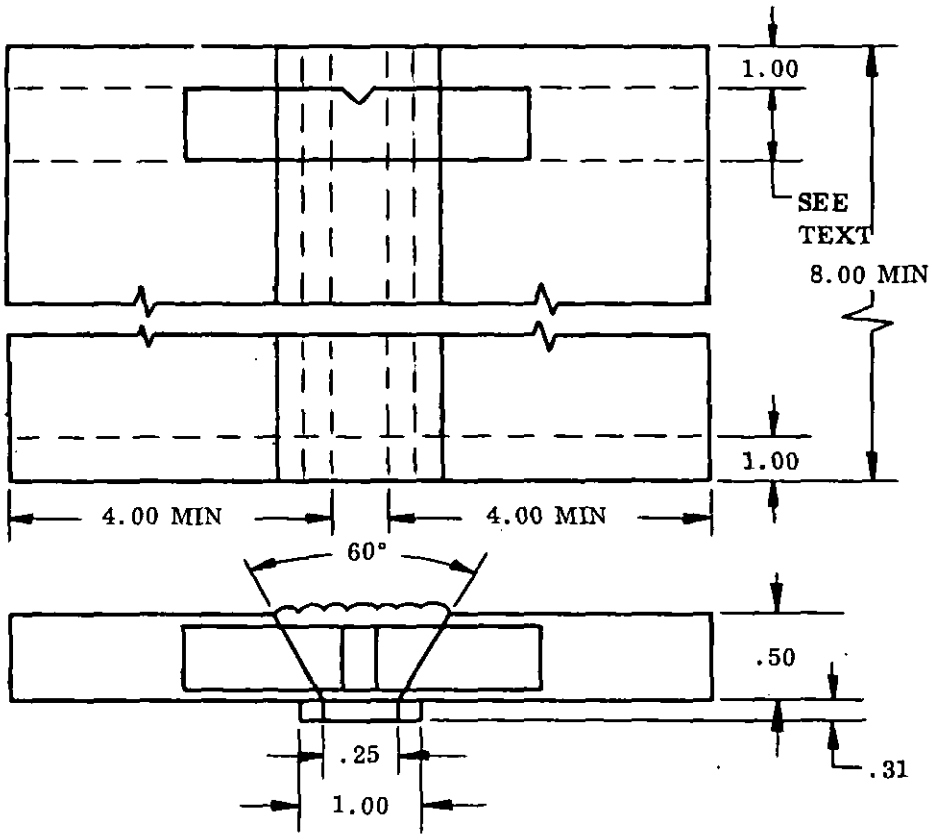


TYPICAL PASS SEQUENCE

DIMENSIONS FOR TENSILE TEST SPECIMENS			
DIMENSIONS		SIZE - ELECTRODES	
		1/8	5/32, 3/16 and 1/4
PLATE-LENGTH	(L)	9-3/4	14-3/4
PLATE-WIDTH	(W)	5	5
PLATE-THICKNESS	(T)	1/2	3/4
BACKING STRIP-THICKNESS	(B)	3/16	1/4
GROOVE WELD OPENING	(R)	1/4	1/2
TENSILE SPECIMEN-DIAMETER	(D)	0.252 ± 0.005	0.505 ± 0.010
TENSILE SPECIMEN-LENGTH	(A)	3	5-1/2

FIGURE 4. ELECTRODE TEST PLATES

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TEST PLATE FOR CHECKING IMPACT PROPERTIES OF WELD DEPOSITS.

BACKING STRIP SHALL BE OF SAME NOMINAL COMPOSITION AS BASE METAL.

DIMENSIONS FOR IMPACT TEST SPECIMENS SHALL CONFORM TO ASTM A370, FIGURE 11 OF TYPE A, CHARPY V-NOTCH.

DIMENSIONS IN INCHES.

FIGURE 5. TEST PLATES FOR IMPACT TEST SPECIMEN

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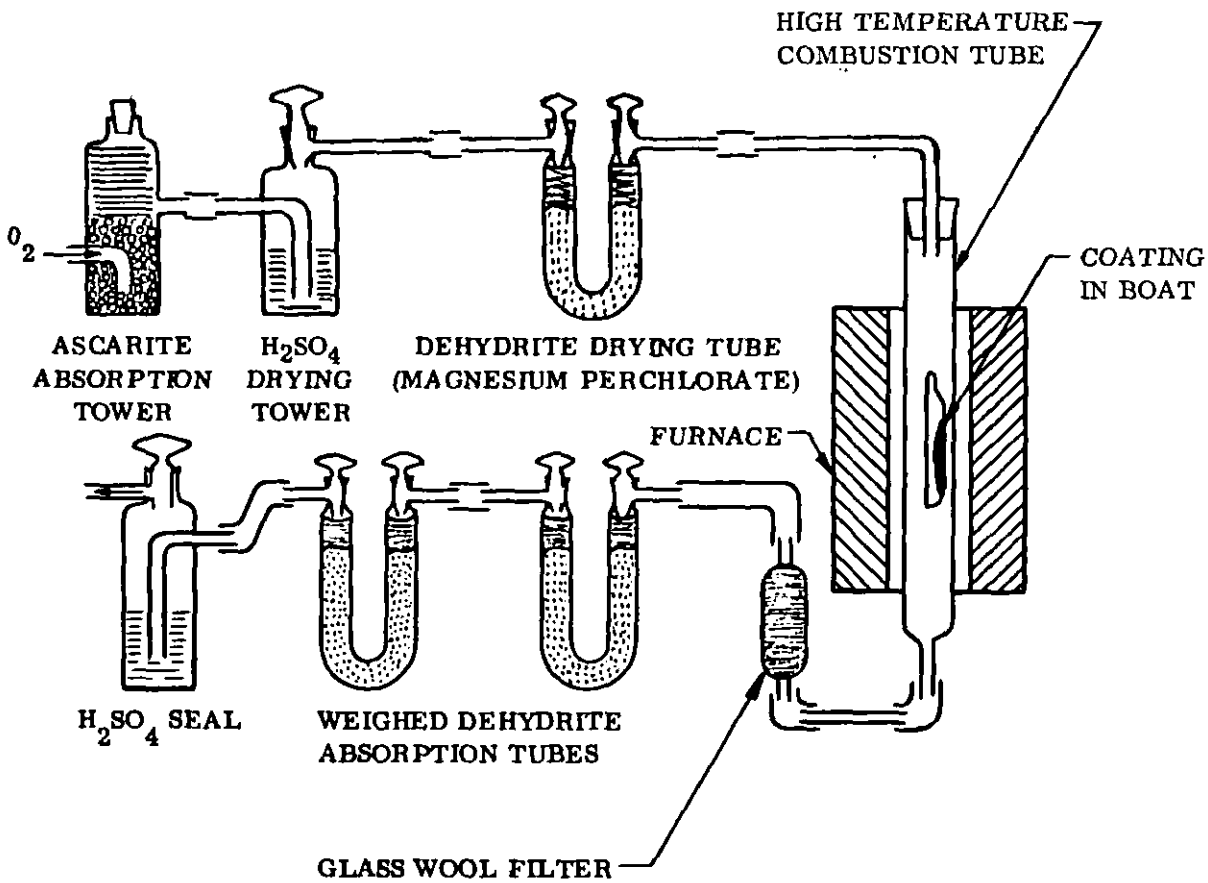


FIGURE 6. SCHEMATIC DRAWING OF TRAIN FOR WATER DETERMINATION

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CONTRACT NUMBER

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   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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